Anja Conev

anjaconev.github.io | a.conev@imperial.ac.uk | Google Scholar

Interests

I am passionate about applying and teaching computational techniques and artificial intelligence in the context of computational biology and bioinformatics.

My PhD thesis focused on algorithmic work in the field of computational structural biology. Over the past year I have worked in an interdisciplinary environment of a more general bioinformatics focusing on genetic variant modelling.

Throughout my studies and postdoctoral work I have been devoted to teaching and mentoring students at both undergraduate and graduate levels.

Education

2019 - 2024 PhD in Computer Science (GPA: 4.0/4.0)
thesis: *Adapting Learning and Search Algorithms to Handle Protein Structural Data*
advisor: Dr. Lydia Kavraki
Rice University
Houston, TX

2014 - 2019 B.S. in Electrical Engineering and Computer Technology (GPA: 9.1/10)
University of Belgrade
Belgrade, Serbia

Professional Experience

CISBIO *Imperial College London, London, UK*	Post-Doctoral Researcher; Bioinformatician *mentor: Prof. Michael Sternberg*	Oct 2024 - Now
Computer Science Department *Rice University, Houston, TX, USA*	Teaching Assistant *mentor: Prof. Devika Subramanian*	2020 - 2024
KavrakiLab *Rice University, Houston, TX, USA*	Graduate Student *mentor: Prof. Lydia Kavraki*	2019 - 2024
eFront *Belgrade, Serbia*	Junior Full-Stack Developer	2018 - 2019
Electrical Engineering School *University of Belgrade, Serbia*	Student Demonstrator	2018 - 2019

Teaching/Mentorship

Teaching assistant at Imperial College London 2025-2026
Department of Life Sciences, Imperial College London
Experience designing and runing lectures on the following topics. Lectures included theorerical background, coding sessions, tutorials, and grading for masters and undergraduate students with diverse academic backgrounds

- Computational Omics - RNAseq Practical (Undergraduate lebe;)
- Code version control with GitHub (MRes level)
- Database and web development with MySQL and Django frameworks (MRes level)
- Python introductory coding sessions (Undergraduate level)

Mentoring interning students at CISBIO 2025-2026
CISBIO, Department of Life Sciences, Imperial College London
Mentored masters students on their interdisciplinary projects in the field of structural computational biology and applied AI.

- *students: Eleanor Stevens, Rick Lee, Zhi Yee Liang, Fidan Maharramova, Swadha Gupta*

Teaching assistant at Rice University 2020-2024
Computer Science Department, Rice University
Experience running recitation sessions, constructing quiz questions, grading, running office hours, writing blogs and mentoring students for the final project.

- COMP 600 (Graduate Seminar in Computer Science) - Spring 2024
- COMP 557 (Artificial Intelligence) *lead teaching assistent*- Fall 2020, Fall 2021
- COMP 540 (Statistical Machine Learning) - Spring 2021

Mentoring interning students at KavrakiLab 2021-2024
KavrakiLab, Computer Science Department, Rice University
Experience guiding students who joined KavrakiLab and worked on projects in the field of structural computational biology and applied machine learning.

- Teresa Zhou (Rice University) - Spring/Summer 2024
- Jaila Lewis (University of Houston) - Summer 2022
- Aleksandar Gavric (University of Belgrade) - Summer 2021
- Davyd Fridman (Rice University) - Summer 2021
- Nonso Chukwurah (Rice University) - Spring 2021

Student demonstrator at University of Belgrade 2018
School of Electrical Engineering, University of Belgrade

Publications

Conev, Anja et al. (Dec. 2025). "3DSeqCheck: A Web-based Tool for Verifying Sequence Consistency Between a 3D Structure File and the Corresponding UniProt Entry". In: *Journal of Molecular Biology*, p. 169620. ISSN: 0022-2836. DOI: 10.1016/j.jmb.2025.169620. URL: http://dx.doi.org/10.1016/j.jmb.2025.169620.

Conev, Anja et al. (Aug. 2025). “DINC-ensemble: A web server for docking large ligands incrementally to an ensemble of receptor conformations”. In: *Journal of Molecular Biology* 437.15, p. 169163. ISSN: 0022-2836. DOI: 10.1016/j.jmb.2025.169163. URL: http://dx.doi.org/10.1016/j.jmb.2025.169163.

Slone, Jared K. et al. (Jan. 2025). “TCR-pMHC Binding Specificity Prediction From Structure Using Graph Neural Networks”. In: *IEEE Transactions on Computational Biology and Bioinformatics* 22.1, 171–179. ISSN: 2998-4165. DOI: 10.1109/tcbbio.2024.3504235. URL: http://dx.doi.org/10.1109/TCBBIO.2024.3504235.

Tsitsa, Ifigenia et al. (Dec. 2025). “The aging of the AlphaFold database”. In: *Nature Structural and Molecular Biology* 32.12, 2374–2376. ISSN: 1545-9985. DOI: 10.1038/s41594-025-01725-z. URL: http://dx.doi.org/10.1038/s41594-025-01725-z.

Conev, Anja et al. (2024). “HLAEquity: Examining biases in pan-allele peptide-HLA binding predictors”. In: *iScience* 27.1, p. 108613. ISSN: 2589-0042. DOI: https://doi.org/10.1016/j.isci.2023.108613. URL: https://www.sciencedirect.com/science/article/pii/S2589004223026901.

Conev, Anja et al. (2023). “EnGens: a computational framework for generation and analysis of representative protein conformational ensembles”. In: *Briefings in Bioinformatics* 24.4, bbad242. DOI: 10.1093/bib/bbad242.

Conev, Anja et al. (June 2022). “3pHLA-score improves structure-based peptide-HLA binding affinity prediction”. In: *Scientific Reports* 12.1. Featured in the collection: ”Top 100 in Chemistry - 2022” https://www.nature.com/collections/jigiddbdcf. DOI: 10.1038/s41598-022-14526-x. URL: https://doi.org/10.1038/s41598-022-14526-x.

Rigo, Mauricio Menegatti et al. (July 2022). “SARS-Arena: Sequence and Structure-Guided Selection of Conserved Peptides from SARS-related Coronaviruses for Novel Vaccine Development”. In: *Frontiers in Immunology* 13. DOI: 10.3389/fimmu.2022.931155. URL: https://doi.org/10.3389/fimmu.2022.931155.

Antunes, Dinler A. et al. (Nov. 2020). “HLA-Arena: A Customizable Environment for the Structural Modeling and Analysis of Peptide-HLA Complexes for Cancer Immunotherapy”. In: *JCO Clinical Cancer Informatics* 4, pp. 623–636. DOI: 10.1200/cci.19.00123. URL: https://doi.org/10.1200/cci.19.00123.

Conev, Anja et al. (Dec. 2020). “Machine Learning-Guided Three-Dimensional Printing of Tissue Engineering Scaffolds”. In: *Tissue Engineering Part A* 26.23-24, pp. 1359–1368. DOI: 10.1089/ten.tea.2020.0191. URL: https://doi.org/10.1089/ten.tea.2020.0191.

Sajkunic, Sanja and Anja Conev (July 2013a). “Stage and Sex Structure of Red-Backed Shrike (Lanius collurio Linnaeus, 1758) Nesting Groups on the Territory of Petnica village”. In: *Petnica Science Center Students’ Projects* 72, pp. 250–253. URL: https://esveske.github.io/pdf/2013/BIO1306.pdf.

– (July 2013b). “The Effect of Lead on Fitness Components in D. subobscura”. In: *Petnica Science Center Students’ Projects* 72, pp. 223–231. URL: https://esveske.github.io/pdf/2013/BIO1306.pdf.

Awards and Fellowships

Ken Kennedy-HPE Cray Fellowship 2023-2024

Ken Kennedy Institute 2023/24

Industry-basec scholarship to further research pursuits, attend conferences, travel, and develop networking relationships in industry.

MolSSI Software Fellowship 2023-2024

MolSSI 2023/24

A 12-month Fellowship from the MolSSI institute for pursuing the development of software infrastructure, middleware, and frameworks that benefit the broader field of computational molecular sciences, including biomolecular and macromolecular simulation, quantum chemistry, and materials science.

Best research presentation May 2023

COMP600, 2023

Award for the best research presentation in the Computer Science PhD cohort at Rice University in the Graduate Research Seminar COMP600.

ITN travel award Sep 2022

ITCR 2022

Travel award to visit the NCI Informatics Technology for Cancer Research (ITCR) conference and participate in a training workshop.

Rice Datathon 2022 - Second Place in the Bill.com Challenge Jan 2022

Rice Datathon 2022

For our work titled *Insight into Connection* on link prediction task with a graph neural network approach.

Poster Presentation Session: Potential Impact Award Oct 2020

2020 Ken Kennedy Institute Data Science Conference

For the poster titled: "Combining Structure and Sequence Data to Predict Peptide-HLA Binding Affinity"

"Dositeja" scholarship for studying abroad 2019-2021

Ministry of Youth and Sport, Republic of Serbia

"Dositeja" is a scholarship awarded by the Fund for Young Talents of the Republic of Serbia to talented and successful students.

Conferences

- **London Protein Design Day**, June 2026
- **ISCB-UK 2025**, April 2026
  - Poster presentation *Exploring discrepancies of structure-based resources such as AlphaFoldDB in relation to the evolving UniProt entries*
- **UK-QSAR Meeting**, 2024/2025
  - Oxford, Fall 2024
  - London, Spring 2024
  - Cambridge, Fall 2025
- **Future Leaders Summit**, April 2024
  Responsible Data Science and AI, MIDAS, University of Michigan
- **ITCR 2023**, Sept 2023
  Data for the Common Good, University of Chicago, Chicago, IL
  - Research presentation *PROTEAN-CR: A proteomics toolkit for ensemble analysis in cancer research*
- **ACM-BCB 2023**, Sept 2023
  Houston, TX

- **CECAM-PsiK**, June 2023
  Freie Universitat, Berlin, Germany
- **CRA-WP Grad Cohort for Women**, April 2023
  San Francisco, SF, US
- **27th SCSB Annual Symposium**, April 2023
  Galveston, TX, US
- **ITCR 2022**, September 2022
  St Louis, WA, US

Open-source projects

3DSeqCheck

3DSeqCheck is a lightweight Python tool to compare UniProt sequences with sequences extracted from 3D structures.

HLAEquity

In our work on HLAEquity we examine allele biases in pan-allele machine learning peptide-HLA binding affinity predictors. We provide an interactive platform for visualizing and assessing dataset and algorithmic bias using a population coverage metric.

EnGens

EnGens is a computational framework for generation and analysis of representative protein conformational ensembles. EnGens performs dimensionality reduction and clustering of the structural datasets to identify representatives from each cluster into a subset of a representative ensemble that can be used for downstream tasks.

SARS-Arena

This work emerged as our response to the SARS-Cov-2 pandemic. Leveraging the resources we previously developed, we tuned our antigen prediction pipelines to fit the purpose of SARS-Cov-2 related antigen discovery.

3pHLA-score

In this work we explored the use of ML models for training a system-specific scoring function for the purpose of scoring the binding energy of peptide ligands to HLA protein receptors. The main contribution includes the novel per-peptide-position training protocol as a proof of concept.

HLA-Arena

HLA-Arena provides an interactive pipeline for structural computational modeling and analysis of the human leukocite antigen (HLA) protein and its interaction with the peptide ligands (antigens).

Training

EMBL-EBI resources in practice 2025 — November 2025
Chemicals in a biological context: metabolites, drugs, and beyond, Virtual

ITN Trainee Workshop 2023 — Sept 2023
Data for the Common Good, University of Chicago, Chicago, IL

MolSSI Software Fellow Bootcamp 2023 — July 2023
MolSSI Institute, Virginia Tech, Blacksburg, VA

Computer Science Student Advancement Program — July - Oct 2018
Kavraki Lab, Rice University, Houston, Texas

Summer School of Machine Learning July 2017
Petnica Science Center and Microsoft Development Center, Serbia

Experimental Chemistry, Biology and Biomedicine research camps 2011 - 2013
Petnica Science Center, Serbia

Professional service and memberships

- Peer-review for journals:
  - Nature Communications
  - Journal of Chemical Physics
  - Journal of Open Source Software
  - Briefings in Bioinformatics
  - Journal of Molecular Biology
  - PCI Genomics
  - Frontiers in Neurology
- Program Committee member for CIBB 2026
- ISCB Member (from 2020)

Skills and hobbies

Programming skills	Python, R, C, C++, Java
Operating systems	Linux, Windows
Communication skill	excellent communication skills gained through my experience with collaborative interdisciplinary research as well as past experiences as a television presenter and organizer, and in a youth theater group
Organizational skills	excellent organisational skills gained through the experience in mentorship, teaching and collaborative projects
Hobbies	guitar, drama, yoga, creative writing